

Kotaro Ikeda

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EDUCATION

The University of Tokyo Tokyo, Japan
BE, Department of Mathematical Engineering and Information Physics, School of Engineering Apr 2022 – Mar 2026

- Planned to graduate in March 2026
- Thesis advisor: Prof. Hiroshi Kori
- (unofficial) GPA: 3.87/4.3
- Dean's Award for Outstanding Research (工学部長賞 (研究優秀)) (Mar 2026)

FUNDINGS

Funai Overseas Scholarship Sept 2026 – Aug 2028

Partial funding for studying abroad.

- \$36,000 per year for living cost
- \$14,000 per year for tuition
- Full coverage of medical insurance

PUBLICATIONS

- Pairwise Optimal Transports for Training All-to-All Flow-Based Condition Transfer Model
Kotaro Ikeda, Masanori Koyama, Jinzhe Zhang, Kohei Hayashi, Kenji Fukumizu
NeurIPS 2025 Poster (Acceptance rate: 24.52%) Dec 2025
<https://neurips.cc/virtual/2025/loc/san-diego/poster/117564>
- Speed-accuracy relations for diffusion models: Wisdom from nonequilibrium thermodynamics and optimal transport
Kotaro Ikeda, Tomoya Uda, Daisuke Okanohara, Sosuke Ito
Physical Review X 15.3 (2025): 031031 (Top Journal in Physics [Impact Factor: 15.7]) June 2025
<https://doi.org/10.1103/x5vj-8jq9>

PROJECTS

Board Game agent development @ May Fes Feb 2025 – May 2025

- Created a reinforcement learning based agent for Blokus™
- Created a website and a real-world competition system for playing with our agents.
- Served as a project leader at May Fes.

Research Experience Seminar on the Universal Principles of Life Apr 2023 – Mar 2024
Small-group seminar by Universal Biology Institute, the University of Tokyo

Advised by Prof. Sosuke Ito

- Constructed a novel analogy between diffusion models and stochastic thermodynamics.
- Theoretically discussed the optimality of noise schedules in diffusion models.
- Published a paper on this topic in a top-physics journal.

EXPERIENCE

Part-time Engineer, Meltly, Inc. Apr 2025 – present
Tokyo, Japan

Teaching Assistant, The University of Tokyo
Tokyo, Japan

Apr 2025 – Aug 2025

Summer Internship, Preferred Networks, Inc.
Tokyo, Japan

Aug 2024 – Feb 2025

Part-time Engineer, Akari, Inc.
Tokyo, Japan

June 2023 – Mar 2024

CONFERENCES IN JAPAN

- Speed-accuracy trade-off for the diffusion models: Wisdom from nonequilibrium thermodynamics and optimal transport
Kotaro Ikeda (presenter), Tomoya Uda, Daisuke Okanohara, Sosuke Ito
The 27th Information-Based Induction Sciences Workshop (IBIS), Saitama, Japan Nov 2024
- GeoBeta Flow Matching
Kotaro Ikeda (presenter), Masanori Koyama, Jinzhe Zhang, Kohei Hayashi, Kenji Fukumizu
The 27th Information-Based Induction Sciences Workshop (IBIS), Saitama, Japan Nov 2024